

REMO: Deterministic Scenario Replay and Modification for ADS Debugging in CARLA

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Summary

In simulation-based testing of autonomous driving systems (ADS), the ability to reproducibly recreate and modify driving scenarios is crucial for testing and debugging system performance. For example, when an ADS fails to navigate a complex urban driving scenario, researchers and engineers need to isolate the contributing factors by systematically modifying scenario elements, such as weather conditions, time of day, street lights, traffic signs, or the presence of other vehicles and pedestrians. However, existing simulation platforms like CARLA (Dosovitskiy et al., 2017) often lack the capability to deterministically replay scenarios with modified parameters while maintaining consistent non-player character (NPC) behaviour.

To address this limitation, we present REMO, a tool that enables deterministic replay and modification of simulation scenarios within CARLA. REMO provides the following key features: * **Deterministic Scenario Replay:** REMO allows users to record and replay any driving scenarios in CARLA while ensuring that all static and dynamic elements, including NPC behaviours, remain consistent across replays, unless intentionally modified. * **Scenario Modification:** Users can modify various scenario elements/parameters, including environmental conditions (e.g. weather, time of day) and the presence of specific actors (e.g. vehicles, pedestrians), without disrupting the deterministic nature of the replay. * **ADS-Agnostic Testing:** REMO supports scenario replay with or without an ego vehicle (i.e. the autonomous vehicle under test), enabling researchers to test different ADS models under identical NPC conditions for fair comparisons. By enabling deterministic replay and flexible scenario modification, REMO facilitates systematic debugging and evaluation of autonomous driving systems.

Statement of Need

Debugging failures in machine learning-enabled autonomous driving systems (ADS) is a complex task that requires the ability to reproducibly recreate and modify driving scenarios (Shin & Pennada, 2024). For example, Arcaini et al. (2022) proposed iterative removal of NPC vehicles to isolate the minimal set of NPCs responsible for triggering a failure, showing that simplified scenarios significantly aid engineers during debugging. However, they focused on NPC removal and did not address the broader need for deterministic scenario replay and modification of other scenario elements (e.g. weather, time of day, buildings). Furthermore, due to the inherent non-determinism of simulation platforms (Chance et al., 2022; Osikowicz et al., 2025), simply re-running a scenario without explicitly recording and replaying it can lead to different outcomes, making it difficult to isolate failure causes. Although CARLA supports record and replay functionality, it lacks the ability to modify scenarios while maintaining deterministic NPC behaviour.

REMO is designed to fill this gap by providing a tool that enables deterministic replay of recorded scenarios with the ability to modify various scenario parameters without affecting NPC behaviour. It allows users to systematically explore how different factors contribute to ADS failures by enabling controlled modifications of the simulation environment. It also

43 supports ADS-agnostic testing by allowing scenario replay without an ego vehicle, facilitating
44 fair comparisons between different ADS models under identical conditions.

45 Overview of the Tool

46 Figure [Figure 1](#) illustrates the overall architecture of REMO. (More to be added later; please
47 mention key components that enable the features described above. This can also be added to
48 the repository README later.)

Overall Architecture of REMO
Figure 1: Overall Architecture of REMO

49 Repository and Installation

50 The tool requires Python 3.8 and Carla version 0.9.15. The source code of REMO, including
51 detailed installation instructions and example scripts, is hosted on GitHub: <https://github.com/RSE-Sheffield/REMO>.
52

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